

HW6 (Chapter 10) for MSE 201:002 Spring 2023

Textbook:

W. D. Callister and D.G. Rethwisch, Materials Science and Engineering: An Introduction, 8th Edition

Chapter 10

- 10.2 (a)** Rewrite the expression for the total free energy change for nucleation (Equation 10.1) for the case of a cubic nucleus of edge length a (instead of a sphere of radius r). Now differentiate this expression with respect to a (per Equation 10.2) and solve for both the critical cube edge length, a^* , and also ΔG^* .
- (b)** Is ΔG^* greater for a cube or a sphere? Why?

- 10.4 (a)** For the solidification of iron, calculate the critical radius r^* and the activation free energy ΔG^* if nucleation is homogeneous. Values for the latent heat of fusion and surface free energy are $-1.85 \times 10^9 \text{ J/m}^3$ and 0.204 J/m^2 , respectively. Use the supercooling value found in Table 10.1.
- (b)** Now calculate the number of atoms found in a nucleus of critical size. Assume a lattice parameter of 0.292 nm for solid iron at its melting temperature.

- 10.5 (a)** Assume for the solidification of iron (Problem 10.4) that nucleation is homogeneous, and the number of stable nuclei is 10^6 nuclei per cubic meter. Calculate the critical radius and the number of stable nuclei that exist at the following degrees of supercooling: 200 K and 300 K.

10.9 The kinetics of the austenite-to-pearlite transformation obey the Avrami relationship. Using the fraction transformed–time data given here, determine the total time required for 95% of the austenite to transform to pearlite:

<i>Fraction Transformed</i>	<i>Time (s)</i>
0.2	12.6
0.8	28.2

10.12 Determine values for the constants n and k (Equation 10.17) for the recrystallization of copper (Figure 10.11) at 102°C.